

Course Level: Undergraduate Open Elective (B. Tech)

BL: 2 Understand

CO: Explain the fundamental principles of nutrition.

Module 2: Digestion & Absorption of Lipids, Vitamins & Minerals

1. Digestion of Lipids

Mechanical disruption: mouth & stomach → lipid particles → emulsions.

Key step: emulsification into fine oil droplets in water.

Stabilization: by membrane lipids, denatured proteins, polysaccharides, phospholipids, cholesterol.

Enzymes involved:

Gastric lipase (GL).

Pancreatic lipase (PL).

Carboxyl ester hydrolase, phospholipase A2.

Role of bile salts: act as surfactants, dispersing hydrophobic lipids in aqueous environment.

2. Absorption of Lipids

Lipolysis products enter lumen as vesicles, mixed vesicles, monomers.

Bile salts: convert emulsions → unilamellar/mixed vesicles.

Barriers: aqueous layer must be crossed.

Mechanisms of entry into enterocytes:

Non-ionic diffusion (uncharged fatty acids).

Collision & incorporation into cell membrane.

Carrier-mediated transport (fatty acid translocase).

3. Absorption of Vitamins

Water-soluble vitamins (WSV): Absorbed mainly in small intestine; some produced by gut microbes in large intestine.

Specific carrier-mediated transport (often sodium-dependent).

Hydrolyzed in stomach → absorbed in duodenum/jejunum.

Vitamin B12 absorbed in ileum.

Transported via portal vein → liver (storage/circulation).

Excess excreted in urine.

Require frequent supply (not stored).

Fat-soluble vitamins (FSV): Absorbed like lipids → incorporated into micelles with bile salts.

Packed into chylomicrons → lymphatic system.

Stored in body (long-term).

Efficient absorption requires dietary fat.

4. Absorption of Minerals

General mechanisms:

Paracellular transport: ions diffuse down electrochemical gradient via tight junction pores.

Solvent drag: minerals carried with water through pores.

Transcellular transport: via ion channels/transporters depending on body needs.

Ion channels: present in membranes of excitable cells and organelles.

5. Example: Calcium Absorption

Sources: diet + digestive secretions.

Not all bioavailable (e.g., Ca-oxalate complexes not absorbed).

Small intestine absorbs ~500 mg/day, secretes ~325 mg/day → net ~**175 mg/day**.

Sites:

Duodenum: active transport.

Jejunum & ileum: diffusion.

Controlled by Calbindin (intracellular Ca regulation).

Requires 1,25-dihydroxy Vitamin D.

6. Example: Iron Absorption

Forms:

Heme iron: absorbed more efficiently.

Non-heme iron: absorbed less efficiently.

Overall absorption: 10–20%.

Ferrous (Fe^{2+}) absorbed better than ferric (Fe^{3+}).

Vitamin C helps convert ferric → ferrous.

Non-heme iron absorbed in duodenum via active transport (Mobilferrin).

In blood: ferrous re-oxidized to ferric → transported by transferrin.

Storage:

Ferritin: major storage form.

Hemosiderin: insoluble storage form (small amounts).

7. Site of Nutrient Absorption (Summary)

Carbohydrates: monosaccharides absorbed in small intestine.

Proteins: amino acids & oligopeptides absorbed in small intestine.

Lipids: absorbed in small intestine via micelles/chylomicrons.

Vitamins:

WSV: duodenum/jejunum (B12 in ileum).

FSV: small intestine with lipids.

Minerals: absorbed throughout small intestine, depending on type.